Regional Simplicial Decomposition and Post-Inversion Empirical Bayes Calibration for Zero-Shot and Few-Shot Topological Neurodynamics in Multichannel EEG

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Repository: `https://github.com/RW-101/TRR3-formalization-`

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Abstract

Zero-lag volume conduction and inter-subject baseline shift remain two primary mathematical obstacles to deploying algebraic topological data analysis (TDA) in real-time brain-computer interfaces (BCIs). While Vietoris–Rips filtration over the finite field $\mathbb{F}_2$ induced by Imaginary Coherency (ImCoh) guarantees exact invariance to instantaneous volume conduction ($\Delta\phi = 0$), scaling global 20-channel minimum spanning tree (MST) persistence $L_{\text{tot}}(H_0)$ from small homogeneous benchmarks ($N = 5$, AUC $= 0.764$) to heterogeneous cohorts ($N = 20$, 1,160 epochs) causes severe accuracy degradation (AUC $= 0.650$) and sign inversions in 20% of subjects (4/20). In this paper, we prove that this degradation stems from three mechanisms: (i) zero-lag denominator inflation under high-amplitude standing waves, (ii) sub-epoch phase cancellation across 4-second windows, and (iii) combinatorial spatial MST dilution across anterior artifact channels. We introduce the **TRR 3.0 Stage 7** architecture to resolve all three bottlenecks. First, we formulate Sub-Window Rectified and Lag-Orthogonalized Imaginary Coherency (rLagImCoh), proving that it preserves exact zero-lag isolation (rLagImCoh($\Delta\phi = 0$) $\equiv 0$) while preventing standing-wave denominator suppression. Second, we introduce an anatomical simplicial decomposition partitioning the 10–20 montage into an 8-channel occipito-parietal complex $\mathcal{K}_{\text{post}}$ and a 12-channel anterior reference complex $\mathcal{K}_{\text{ant}}$, reducing 2-simplex boundary complexity by 4.13× (1,140 → 276 triangles). Third, we derive a strictly causal Post-Inversion Empirical Bayes Stein Shrinkage Fusion protocol under Leave-One-Subject-Out Cross-Validation (LOSO-CV) that prevents covariance-inversion gate cancellation. Audited across $N = 20$ subjects (1,160 epochs) from the PhysioNet EEGBCI database, the pure $\mathbb{F}_2$ topological spatial gradient elevates zero-shot (0 s) accuracy from 0.650 to 0.837 (mean within-subject AUC $= 0.927 \pm 0.022$) and eliminates all subject polarity inversions (20/20 consistent directionality, paired Wilcoxon $p = 1.91 \times 10^{-6}$). Under causal prefix calibration, Pure Topological TRR ($\mathbb{F}_2$ homology only, 0% local power) scales monotonically in pooled accuracy to 0.874 at 22 s ($k = 10$) and 0.895 at 60 s, while the Hybrid Spectral-Topological TRR architecture achieves AUC $= 0.869$ at zero-shot (0 s), 0.897 at 8 s ($k = 3$), 0.928 at 22 s ($k = 10$), and 0.954 at 60 s ($d = +2.34$).

Keywords: Relational Resolution Theory (TRR 3.0), Persistent Homology over $\mathbb{F}_2$, Imaginary Coherency, Volume Conduction Invariance, Empirical Bayes Stein Shrinkage, Zero-Shot BCI Calibration, Multichannel EEG.

1 Introduction

Topological Data Analysis (TDA) provides a coordinate-free mathematical framework for extracting multiscale relational invariants from point clouds and weighted networks. However, generic TDA is agnostic to the biophysics of electromagnetic field propagation in neural tissue. **Relational Resolution Theory 3.0 (TRR 3.0)** [1] bridges this gap by embedding exact simplicial homology over the prime field $\mathbb{F}_2 = \{0, 1\}$ within a physically constrained phase-lagged metric space designed specifically for multichannel electroencephalography (EEG). By constructing the Vietoris–Rips filtration over Imaginary Coherency (ImCoh) [2], TRR 3.0 enforces an exact physical boundary condition: any instantaneous linear mixing of cortical dipoles ($\Delta\phi = 0 \pmod{\pi}$) yields zero imaginary cross-spectrum and maps to the metric ceiling $D(u, v) = 1.0$, guaranteeing immunity against volume-conduction short-circuiting.

Despite this formal guarantee, scaling head-wide topological invariants across diverse human populations reveals a critical bottleneck. On a 5-subject benchmark (`S001-S005`, 290 epochs) from the PhysioNet EEG Motor Movement/Imagery Database (EEGBCI) [4], uncalibrated 20-channel zero-dimensional persistence $L_{\text{tot}}(H_0)$ separates resting-state eyes-open (EO) from eyes-closed (EC) neurodynamics with ROC-AUC $= 0.764$, rising to 0.788 after 60 seconds of Empirical Bayes calibration. However, when the cohort is expanded fourfold to $N = 20$ subjects (`S001-S020`, 1,160 epochs), global uncalibrated accuracy drops to AUC $= 0.650$ ($d = -0.60$), and 60-second univariate calibration reaches only 0.654.

In this study, we prove mathematically and demonstrate empirically that this degradation is not a limitation of $\mathbb{F}_2$ persistent homology, but rather a conse-

quence of three pre-filtration and spatial aggregation bottlenecks. We then present the complete **TRR 3.0 Stage 7** mathematical architecture that resolves these bottlenecks and achieves state-of-the-art zero-shot and few-shot classification across all 20/20 subjects.

2 Mathematical Diagnosis of Global Degradation on Heterogeneous Cohorts

Let $\mathbf{x}(t) \in \mathbb{R}^{N_{\text{ch}}}$ ($N_{\text{ch}} = 20$) denote the average-referenced, bandpass-filtered EEG time series over a 4-second epoch of $W = 640$ samples ($f_s = 160$ Hz), and let $z_u(t) = x_u(t) + i\,\mathcal{H}[x_u](t)$ denote the analytic signal obtained via the discrete Hilbert transform $\mathcal{H}$.

2.1 Zero-Lag Denominator Inflation

In standard 4-second Imaginary Coherency [2], the functional coupling between channels u and v is evaluated as:

$$\text{ImCoh}_{uv} = \frac{|\,\text{Im}(S_{uv})|}{\sqrt{P_u P_v}}, \quad S_{uv} = \frac{1}{W}\sum_{t=1}^{W} z_u(t) z_v^*(t), \tag{1}$$

where $P_u = S_{uu} = \text{Re}(S_{uu})$ is the total narrowband power. During eyes-closed (EC) alpha synchronization (8–13 Hz), the analytic signal at electrodes u, v is a superposition of a dominant zero-lag standing-wave projection $A_0(t)$ ($\Delta\phi = 0$, volume-conducted from radial occipito-parietal dipoles) and true phase-lagged travelling cortical waves $\epsilon_u(t), \epsilon_v(t)$ ($\Delta\phi \neq 0$):

$$z_u(t) = A_0(t) + \epsilon_u(t), \quad z_v(t) = A_0(t) + \epsilon_v(t). \tag{2}$$

Because $\text{Im}(A_0(t)A_0^*(t)) \equiv 0$, the numerator $|\,\text{Im}(S_{uv})|$ scales exclusively with the lagged interaction $|\,\text{Im}(\langle\epsilon_u\epsilon_v^*\rangle)|$. In contrast, the denominator $\sqrt{P_u P_v} \approx |A_0|^2 + |\epsilon|^2$ scales linearly with the zero-lag standing-wave power $|A_0|^2$. In subjects whose Berger reaction is dominated by a high-amplitude radial standing wave, $|A_0|^2$ increases by a factor of 5–10× during EC, inflating the denominator of Eq. (1) faster than the numerator. Consequently, ImCoh_{uv} paradoxically decreases during EC, and the metric distance $D(u,v) = 1 - \text{ImCoh}_{uv}$ shifts in the inverted direction ($d > 0$ rather than $d < 0$). As documented in Table 1, exactly 4/20 subjects (`S006`, `S007`, `S012`, `S015`) exhibit inverted effect signs ($d > 0$) under global 4-second ImCoh.

2.2 Sub-Epoch Phase Cancellation

Cortical phase-lead relationships are metastable, reversing direction every 300–800 ms ($\sin\Delta\phi(t) > 0 \leftrightarrow \sin\Delta\phi(t) < 0$). Linearly averaging $z_u(t)z_v^*(t)$ over a long $W = 4$ s window (≈ 40 alpha cycles) *before* taking the modulus $|\,\text{Im}(\cdot)|$ results in destructive algebraic cancellation between positive and negative phase-lag episodes.

Table 1: Per-subject diagnostic audit across $N = 20$ subjects (`S001-S020`, 58 epochs/subject). Global 4s ImCoh suffers from polarity inversions ($d > 0$) in 4/20 subjects (†), whereas Stage 7 Posterior Spectral-Topological filtration achieves 20/20 consistent directionality ($d < 0$) and raises mean within-subject AUC from 0.652 to 0.944.

Subject	Old Global H_0 d	Old Global AUC	Stage 7 SpecTopo d	Stage 7 SpecTopo AUC
S001	−0.89	0.734	−8.32	1.000
S002	−1.41	0.847	−2.67	0.981
S003	−0.73	0.653	−4.65	1.000
S004	−3.53	0.993	−5.73	1.000
S005	−0.84	0.690	−1.86	0.920
S006	+0.08†	0.477	−1.33	0.828
S007	+0.60†	0.334	−1.36	0.838
S008	−1.26	0.797	−2.36	0.917
S009	−0.53	0.630	−4.19	0.994
S010	−1.65	0.894	−2.86	0.949
S011	−1.41	0.817	−2.53	0.951
S012	+0.69†	0.332	−3.16	0.996
S013	−0.42	0.592	−2.37	0.989
S014	−1.06	0.757	−3.71	1.000
S015	+2.14†	0.065	−0.97	0.774
S016	−0.56	0.648	−3.61	0.998
S017	−2.42	0.988	−6.13	1.000
S018	−0.35	0.593	−5.96	1.000
S019	−0.39	0.615	−0.98	0.741
S020	−0.52	0.591	−5.11	1.000
Summary	**4/20 Inverted (†)**	**Mean: 0.652**	**0/20 Inverted**	**Mean: 0.944**

2.3 Combinatorial Spatial MST Dilution

Zero-dimensional total persistence $L_{\text{tot}}(H_0)$ over $N_{\text{ch}} = 20$ vertices equals the sum of the $N_{\text{ch}} - 1 = 19$ edge weights of the Minimum Spanning Tree (MST). On the canonical 10–20 montage, posterior visual-parietal cortex is covered by $N_{\text{post}} = 8$ electrodes ($\mathcal{V}_{\text{post}} = \{\text{P7, P3, Pz, P4, P8, O1, Oz, O2}\}$). Any acyclic forest on $\mathcal{V}_{\text{post}}$ contains at most $N_{\text{post}} - 1 = 7$ edges. Therefore, at least $19 - 7 = 12$ edges (63.2% of the global sum $L_{\text{tot}}(H_0)$) must connect the $N_{\text{ant}} = 12$ fronto-central and temporal electrodes ($\mathcal{V}_{\text{ant}}$), where ocular (EOG) and muscular (EMG) noise dominate.

3 The TRR 3.0 Stage 7 Mathematical Architecture

3.1 Sub-Window Rectified Lag-Orthogonalized Imaginary Coherency (rLagImCoh)

Each 4-second epoch is partitioned into $M = 7$ overlapping sub-windows of length $L_{\text{sub}} = 1.0$ s (160 samples) with step 0.5 s (80 samples), tapered by a Hann window $w(t)$. Let $S_{uv}^{(m)} = \frac{1}{L_{\text{sub}}}\sum_t w^2(t) z_u^{(m)}(t) z_v^{(m)*}(t)$ be the local cross-spectrum in sub-window m, and $P_u^{(m)} = \text{Re}(S_{uu}^{(m)})$. We define:

$$\text{rLagImCoh}_{uv} = \frac{1}{M}\sum_{m=1}^{M} \text{clip}_{[0,1]}\left(\frac{\left|\text{Im}\left(S_{uv}^{(m)}\right)\right|}{\sqrt{\max\left(\epsilon_{\text{tol}},\, P_u^{(m)}P_v^{(m)} - \gamma\left[\text{Re}\left(S_{uv}^{(m)}\right)\right]^2\right)}}\right), \tag{3}$$

with Pascual-Marqui lag-orthogonalization parameter $\gamma = 0.85$ [3] and $\epsilon_{\text{tol}} = 10^{-15}$. The induced functional distance matrix is $D_{\text{rLag}}(u,v) = 1 - \text{rLagImCoh}_{uv}$ for $u \neq v$ and $D_{\text{rLag}}(u,u) = 0$.

Proposition 1 (Strict Zero-Lag Volume Conduction Isolation)**.** *For any instantaneous linear mixture of cortical sources ($\Delta\phi_{uv}(t) \equiv 0 \pmod{\pi}$),* $\text{rLagImCoh}_{uv} = 0$ *and* $D_{\text{rLag}}(u,v) = 1.0$ *for all* $u \neq v$.

Proof. If $\Delta\phi_{uv}(t) \equiv 0 \pmod{\pi}$, then $z_u^{(m)}(t) z_v^{(m)*}(t) \in \mathbb{R}$ for all t, so $\mathrm{Im}(S_{uv}^{(m)}) = 0$ in every sub-window $m \in \{1, \ldots, M\}$. By the Cauchy–Schwarz inequality, $[\mathrm{Re}(S_{uv}^{(m)})]^2 \le |S_{uv}^{(m)}|^2 \le P_u^{(m)} P_v^{(m)}$. Since $\gamma = 0.85 < 1$, the radicand in the denominator of Eq. (3) satisfies $P_u^{(m)} P_v^{(m)} - \gamma [\mathrm{Re}(S_{uv}^{(m)})]^2 \ge (1-\gamma) P_u^{(m)} P_v^{(m)}$, and the $\max(\epsilon_{\mathrm{tol}}, \cdot)$ operator guarantees a strictly positive denominator $\ge \sqrt{\epsilon_{\mathrm{tol}}} > 0$. Thus every summand in Eq. (3) is identically zero. □

3.2 Anatomical Simplicial Decomposition over $\mathbb{F}_2$ and Combinatorial Speedup

We partition the vertex set $\mathcal{V}$ ($|\mathcal{V}| = 20$) into two disjoint anatomical sub-complexes:

$$\mathcal{V}_{\mathrm{post}} = \{\mathrm{P7}, \mathrm{P3}, \mathrm{Pz}, \mathrm{P4}, \mathrm{P8}, \mathrm{O1}, \mathrm{Oz}, \mathrm{O2}\} \quad (N_{\mathrm{post}} = 8), \tag{4}$$

$$\mathcal{V}_{\mathrm{ant}} = \{\mathrm{Fp1}, \mathrm{Fp2}, \mathrm{F7}, \mathrm{F3}, \mathrm{Fz}, \mathrm{F4}, \mathrm{F8}, \mathrm{T7}, \mathrm{C3}, \mathrm{Cz}, \mathrm{C4}, \mathrm{T8}\} \quad (N_{\mathrm{ant}} = 12). \tag{5}$$

For each sub-complex $\mathcal{K}_r$ ($r \in \{\mathrm{post}, \mathrm{ant}\}$), the Vietoris–Rips filtration $\mathrm{VR}_\epsilon(\mathcal{V}_r, D_{\mathrm{rLag}})$ orders the 1-simplices (edges) $\mathcal{E}_r = \{e_1, \ldots, e_{|\mathcal{E}_r|}\}$ by weight $w(e) = D_{\mathrm{rLag}}(u, v)$ and the 2-simplices (triangles) $\mathcal{T}_r = \{\tau_1, \ldots, \tau_{|\mathcal{T}_r|}\}$ by diameter $w(\tau) = \max(D_{\mathrm{rLag}}(u, v), D_{\mathrm{rLag}}(v, w), D_{\mathrm{rLag}}(u, w))$.

Let $\partial_1 \in \mathbb{F}_2^{|\mathcal{V}_r| \times |\mathcal{E}_r|}$ and $\partial_2 \in \mathbb{F}_2^{|\mathcal{E}_r| \times |\mathcal{T}_r|}$ denote the simplicial boundary matrices over $\mathbb{F}_2$ (where addition is bitwise XOR $\oplus$). Standard left-to-right column reduction yields pivot pairs (i, j) defining the zero- and one-dimensional persistence invariants:

$$L_{\mathrm{tot}}(H_0 \mid \mathcal{K}_r) = \sum_{v \in \mathrm{Piv}(\partial_1)} w(e_{m_1(v)}), \quad \bar{H}_0(\mathcal{K}_r) = \frac{L_{\mathrm{tot}}(H_0 \mid \mathcal{K}_r)}{N_r - 1}, \tag{6}$$

$$L_{\max}(H_1 \mid \mathcal{K}_r) = \max_{i \in Z_1(\mathcal{K}_r)} \big(w_{\mathrm{death}}(e_i) - w(e_i)\big), \tag{7}$$

$$L_{\mathrm{tot}}(H_1 \mid \mathcal{K}_r) = \sum_{i \in Z_1(\mathcal{K}_r)} \big(w_{\mathrm{death}}(e_i) - w(e_i)\big), \tag{8}$$

where $Z_1(\mathcal{K}_r) = \{i \in \{1, \ldots, |\mathcal{E}_r|\} : \mathrm{piv}(\partial_1(:, i)) = -1\}$ indexes cycle-creating edges, and $w_{\mathrm{death}}(e_i) = w(\tau_{m_2(i)})$ if $i \in \mathrm{Piv}(\partial_2)$ or $\max(D_{\mathrm{rLag}})$ otherwise.

Combinatorial Complexity Reduction: Global filtration on $N_{\mathrm{ch}} = 20$ vertices requires reducing $|\mathcal{T}_{\mathrm{glob}}| = \binom{20}{3} = 1{,}140$ triangles and $|\mathcal{E}_{\mathrm{glob}}| = \binom{20}{2} = 190$ edges. Anatomical decomposition requires only $|\mathcal{T}_{\mathrm{post}}| + |\mathcal{T}_{\mathrm{ant}}| = \binom{8}{3} + \binom{12}{3} = 56 + 220 = 276$ triangles and $\binom{8}{2} + \binom{12}{2} = 28 + 66 = 94$ edges, yielding a 4.13× **reduction in 2-simplices** and a 2.02× **reduction in 1-simplices**.

The dimensionless **Zero-Shot Topological Spatial Gradient** is defined as:

$$\Delta_{\mathrm{spatial}} H_0(t) = \bar{H}_0(\mathcal{K}_{\mathrm{post}}, t) - \bar{H}_0(\mathcal{K}_{\mathrm{ant}}, t). \tag{9}$$

3.3 Pure Topological vs. Hybrid Spectral-Topological Feature Spaces

To ensure strict ablation transparency between phase-lagged $\mathbb{F}_2$ homology and local band power, we construct two distinct feature representations across subbands α (8–13 Hz), α_1 (7.5–10.5 Hz), α_2 (10.0–13.5 Hz), β (14.0–26.0 Hz), and θ (4.0–7.5 Hz):

1. **Pure Topological TRR ($\mathbb{F}_2$ only, 0% Local Power):** The 1D shift-invariant anchor is $\mathbf{x}_{\mathrm{inv}}^{\mathrm{pure}}(t) = [\Delta_{\mathrm{spatial}} H_0^{(\alpha)}(t)] \in \mathbb{R}^1$, and the 7D calibratable topological vector is:

$$\begin{aligned} \mathbf{x}_{\mathrm{cal}}^{\mathrm{pure}}(t) = \Big[& \Delta_{\mathrm{spatial}} H_0^{(\alpha)}, \\ & \bar{H}_0^{(\alpha_1)}(\mathcal{K}_{\mathrm{post}}) - \bar{H}_0^{(\alpha)}(\mathcal{K}_{\mathrm{ant}}), \\ & \bar{H}_0^{(\alpha_2)}(\mathcal{K}_{\mathrm{post}}) - \bar{H}_0^{(\alpha)}(\mathcal{K}_{\mathrm{ant}}), \\ & \bar{H}_0^{(\alpha)}(\mathcal{K}_{\mathrm{post}}) - \bar{H}_0^{(\beta)}(\mathcal{K}_{\mathrm{post}}), \\ & \bar{H}_0^{(\alpha)}(\mathcal{K}_{\mathrm{post}}),\ L_{\max}(H_1^{(\alpha)} \mid \mathcal{K}_{\mathrm{post}}), \\ & L_{\mathrm{tot}}(H_1^{(\alpha)} \mid \mathcal{K}_{\mathrm{post}}) \Big]^T \in \mathbb{R}^7. \end{aligned} \tag{10}$$

2. **Hybrid SpecTopo TRR (Topology + Regional Band-Power Ratios):** Let $P_b(\mathcal{K}_r, t)$ denote the mean narrowband variance across channels in region r and band b. We define two scale-free log-spectral ratios and a coupled spectral-topological index:

$$\mathrm{rel_}\alpha_{\mathrm{post}}(t) = \ln \frac{P_\alpha(\mathcal{K}_{\mathrm{post}}, t) + \epsilon_{\mathrm{tol}}}{P_\beta(\mathcal{K}_{\mathrm{post}}, t) + P_\theta(\mathcal{K}_{\mathrm{post}}, t) + \epsilon_{\mathrm{tol}}}, \tag{11}$$

$$\mathrm{ratio}_{\mathrm{post/ant}}(t) = \ln \frac{P_\alpha(\mathcal{K}_{\mathrm{post}}, t) + \epsilon_{\mathrm{tol}}}{P_\alpha(\mathcal{K}_{\mathrm{ant}}, t) + \epsilon_{\mathrm{tol}}}, \tag{12}$$

$$\mathrm{SpecTopo}(t) = \bar{H}_0^{(\alpha)}(\mathcal{K}_{\mathrm{post}}, t) \cdot \exp\big(-0.5\, \mathrm{rel_}\alpha_{\mathrm{post}}(t)\big). \tag{13}$$

The Hybrid zero-shot anchor is $\mathbf{x}_{\mathrm{inv}}^{\mathrm{hyb}}(t) = [\Delta_{\mathrm{spatial}} H_0^{(\alpha)}, \mathrm{rel_}\alpha_{\mathrm{post}}, \mathrm{ratio}_{\mathrm{post/ant}}, \mathrm{SpecTopo}]^T \in \mathbb{R}^4$, and the full Hybrid calibratable vector is $\mathbf{x}_{\mathrm{cal}}^{\mathrm{hyb}}(t) = [\mathbf{x}_{\mathrm{cal}}^{\mathrm{pure}}(t)^T, \mathrm{rel_}\alpha_{\mathrm{post}}, \mathrm{ratio}_{\mathrm{post/ant}}, \mathrm{SpecTopo}]^T \in \mathbb{R}^{10}$.

3.4 Post-Inversion Empirical Bayes Stein Shrinkage Fusion

In each fold of Leave-One-Subject-Out Cross-Validation (LOSO-CV), let $\mathcal{D}_{-s}$ denote the $N - 1 = 19$ training subjects and s the held-out target subject. Given a strictly causal calibration prefix of $k \in \{0, 1, 2, 3, 5, 10, 29\}$ resting-state EO epochs $\mathbf{X}_{s,\mathrm{EO}}[0 : k]$, each coordinate $j \in \{1, \ldots, p\}$ of $\mathbf{x}_{\mathrm{cal}}$ is standardized via the Normal-Inverse-Gamma (NIG) conjugate posterior against the training prior $(\mu_{p,j}, \sigma_{p,j}^2)$:

$$\tilde{\mu}_{s,j}(k) = \frac{k\, \hat{\mu}_{s,j}(k) + \tau_0\, \mu_{p,j}}{k + \tau_0}, \tag{14}$$

$$\tilde{\sigma}^2_{s,j}(k) = \frac{\max(0,k-1)\hat{s}^2_{s,j}(k) + \nu_0\,\sigma^2_{p,j} + \frac{k\tau_0}{k+\tau_0}(\hat{\mu}_{s,j}(k) - \mu_{p,j})^2}{\max(0,k-1) + \nu_0}, \quad (15)$$

with fixed hyperparameters $\tau_0 = 3.0, \nu_0 = 5.0$, yielding calibrated feature vectors $\mathbf{z}_s(t \mid k) = (\mathbf{x}_{\mathrm{cal},s}(t) - \tilde{\boldsymbol{\mu}}_s(k)) \oslash \tilde{\boldsymbol{\sigma}}_s(k)$.

Every training subject $j \in \mathcal{D}_{-s}$ undergoes the exact same k-epoch causal prefix calibration to form training matrices $\mathbf{Z}_{\mathrm{tr,EO}}(k)$ and $\mathbf{Z}_{\mathrm{tr,EC}}(k)$. Regularized Fisher Linear Discriminant weights are learned strictly on $\mathcal{D}_{-s}$:

$$\Sigma_{\mathrm{pool}} = \frac{1}{2}\big(\mathrm{Cov}(\mathbf{Z}_{\mathrm{tr,EO}}) + \mathrm{Cov}(\mathbf{Z}_{\mathrm{tr,EC}})\big), \quad (16)$$

$$\Sigma_{\mathrm{reg}} = (1-\lambda)\Sigma_{\mathrm{pool}} + \lambda\,\mathrm{diag}(\Sigma_{\mathrm{pool}}), \quad \lambda = 0.20, \quad (17)$$

$$\mathbf{w} = \Sigma^{-1}_{\mathrm{reg}}(\mathbf{m}_{\mathrm{tr,EC}} - \mathbf{m}_{\mathrm{tr,EO}}), \quad (18)$$

$$\tilde{\mathbf{w}} = \frac{\mathbf{w}}{\sqrt{\mathrm{Var}(\mathbf{Z}_{\mathrm{tr,EO}}\mathbf{w}) + \epsilon_{\mathrm{tol}}}}. \quad (19)$$

Because multiplying feature columns by a Stein weight α_k *prior* to $\Sigma^{-1}_{\mathrm{reg}}$ scales $w_j \propto \alpha_k^{-1}$ and cancels α_k in $\mathbf{z}^T\mathbf{w}$, we project the zero-shot anchor $u_{\mathrm{inv}}(t) = \mathbf{z}_{\mathrm{inv},s}(t)^T\tilde{\mathbf{w}}_{\mathrm{inv}}$ and the calibrated discriminant $u_{\mathrm{cal}}(t \mid k) = \mathbf{z}_s(t \mid k)^T\tilde{\mathbf{w}}_{\mathrm{cal}}(k)$ into unit-variance coordinates independently via Eq. (19), and fuse them *post-inversion*:

$$S_s(t \mid k) = (1-\alpha_k)\,u_{\mathrm{inv}}(t) + \alpha_k\,u_{\mathrm{cal}}(t \mid k), \quad \alpha_k = \frac{k}{k+\tau_0}. \quad (20)$$

4 Experimental Results and Statistical Audit ($N = 20$, 1,160 Epochs)

Table 2 and Figure 1 report the complete experimental audit across $N = 20$ subjects (1,160 epochs):

- **Zero-Shot Pure Topological Breakthrough ($k = 0$):** Restricting filtration to the posterior complex $\mathcal{K}_{\mathrm{post}}$ under rLagImCoh raises pooled zero-shot accuracy from 0.650 to 0.780 (19/20 subjects consistent). Subtracting the anterior reference complex $\mathcal{K}_{\mathrm{ant}}$ via $\Delta_{\mathrm{spatial}}H_0$ elevates pooled zero-shot accuracy to **0.837** (Holdout = 0.819, mean within-subject AUC = $\mathbf{0.927 \pm 0.022}$) and achieves 20/20 subject consistency (two-sided paired Wilcoxon $p = 1.91 \times 10^{-6}$).

- **Analysis of Out-of-Fold LOSO Centering at $k = 0$:** In Table 2, uncentered $\Delta_{\mathrm{spatial}}H_0$ yields Full AUC = 0.837 (Holdout = 0.819), whereas `Pure Topological TRR` at $k = 0$ yields Full AUC = 0.830 (Holdout = 0.812) while maintaining the exact same within-subject accuracy (0.927 ± 0.022). This 0.007 difference arises because $\Delta_{\mathrm{spatial}}H_0$ is naturally centered around physical zero ($\bar{H}_0(\mathcal{K}_{\mathrm{post}}) = \bar{H}_0(\mathcal{K}_{\mathrm{ant}})$), whereas subtracting the leave-one-out training mean $\mu_{i,-s}$ across $N - 1 = 19$ subjects introduces a finite-sample jackknife variance inflation factor of $\frac{N}{N-1} = \frac{20}{19}$.

- **Monotonic Full-Pool Causal Calibration Scaling:** Under Post-Inversion Stein Shrinkage Fusion, `Full Pool AUC` increases strictly monotonically with calibration budget k for both Pure Topological TRR ($0.830 \to 0.831 \to 0.836 \to 0.840 \to 0.850 \to 0.874 \to 0.895$) and Hybrid SpecTopo TRR ($0.869 \to 0.874 \to 0.887 \to 0.897 \to 0.908 \to 0.928 \to 0.954$). Note that at $k = 29$ (60 s) in Pure Topological TRR, Fisher LDA optimizes cross-subject pooled separability (Full AUC = 0.895, $d = +1.01$) by assigning higher weight to H_1 cycle persistence, trading a slight shift in mean within-subject AUC ($0.928 \to 0.915$) for superior global threshold alignment.

5 Conclusion

By resolving zero-lag denominator inflation, sub-epoch phase cancellation, and anterior MST dilution, TRR 3.0 Stage 7 proves that exact $\mathbb{F}_2$ persistent homology operates as a highly accurate, volume-conduction-invariant classifier in both zero-shot (AUC = 0.837–0.869) and rapid few-shot (AUC = 0.897–0.928 within 8–22 s) regimes across heterogeneous multi-subject EEG cohorts.

References

[1] R. Weinberg, "Relational Resolution Theory 3.0: Formal Verification in Lean 4 and Volume-Conduction Invariant Simplicial Persistent Homology over $\mathbb{F}_2$ in Multichannel EEG," *Zenodo Preprint*, DOI: `10.5281/zenodo.23454524`, Repository: `https://github.com/RW-101/TRR3-formalization-`, 2026.

[2] G. Nolte et al., "Identifying true brain interaction from EEG data using the imaginary part of coherency," *Clinical Neurophysiology*, vol. 115, no. 10, pp. 2292–2307, 2004.

[3] R. D. Pascual-Marqui, "Instantaneous and lagged measurements of linear and nonlinear dependence between groups of multivariate time series: frequency decomposition," *arXiv:0711.1455*, 2007.

[4] G. Schalk et al., "BCI2000: a general-purpose brain-computer interface (BCI) system," *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 6, pp. 1034–1043, 2004.

Table 2: Stage 7 Verified Audit across $N = 20$ subjects (1,160 epochs, `S001-S020`) under strict Leave-One-Subject-Out Cross-Validation (LOSO-CV). Balanced Holdout evaluates strictly post-calibration epochs (`EO[10:]` vs. `EC[10:]`, 760 epochs). Hierarchical subject-level significance is evaluated via the two-sided paired Wilcoxon signed-rank test across $N = 20$ independent subjects.

Architecture & Strictly Causal Calibration Budget k ($N = 20$ LOSO-CV)	**Calib Time**	**Cohen's d**	**Full Pool AUC**	**Balanced Holdout AUC**	**Mean Subject AUC (±SE)**	**Subject Wilcoxon p (Pos)**
Part 1: Zero-Shot Simplicial Ablation ($k = 0$, 0 s Calibration, Pure $\mathbb{F}_2$ Topology, Uncentered Physical Threshold)						
1. Stage 6 Global H_0 (Baseline 4s ImCoh, 20 ch)	0 s ($k = 0$)	−0.60	0.650	0.626	0.652 ± 0.051	7.30×10^{-3} (16/20)
2. Pure Topo: Posterior Complex rLagImCoh H_0 (8 ch)	0 s ($k = 0$)	−0.85	0.780	0.764	0.905 ± 0.029	3.81×10^{-6} (19/20)
3. Pure Topo: Spatial Gradient ($\mathcal{K}_{\text{post}} - \mathcal{K}_{\text{ant}}$) H_0	0 s ($k = 0$)	−0.82	**0.837**	**0.819**	$\mathbf{0.927 \pm 0.022}$	$\mathbf{1.91 \times 10^{-6}}$ **(20/20)**
Part 2A: Pure Topological TRR ($\mathbb{F}_2$ Simplicial Homology Only, 0% Spectral Power, Post-Inversion Stein LOSO-CV)						
Pure Topological TRR ($\mathbb{F}_2$ only, $k = 0$)	0 s	+0.82	0.830	0.812	0.927 ± 0.022	1.91×10^{-6} (20/20)
Pure Topological TRR ($\mathbb{F}_2$ only, $k = 1$)	4 s	+0.85	0.831	0.811	0.929 ± 0.021	1.91×10^{-6} (20/20)
Pure Topological TRR ($\mathbb{F}_2$ only, $k = 2$)	6 s	+0.86	0.836	0.815	0.929 ± 0.021	1.91×10^{-6} (20/20)
Pure Topological TRR ($\mathbb{F}_2$ only, $k = 3$)	8 s	+0.87	**0.840**	**0.818**	$\mathbf{0.929 \pm 0.021}$	1.91×10^{-6} (20/20)
Pure Topological TRR ($\mathbb{F}_2$ only, $k = 5$)	12 s	+0.89	0.850	0.827	0.929 ± 0.020	1.91×10^{-6} (20/20)
Pure Topological TRR ($\mathbb{F}_2$ only, $k = 10$)	22 s	+0.88	**0.874**	**0.852**	$\mathbf{0.928 \pm 0.021}$	1.91×10^{-6} (20/20)
Pure Topological TRR ($\mathbb{F}_2$ only, $k = 29$)	60 s	+1.01	**0.895**	N/A (Full)	0.915 ± 0.024	1.91×10^{-6} (20/20)
Part 2B: Hybrid SpecTopo TRR ($\mathbb{F}_2$ Simplicial Homology + Regional Spectral Profile, Post-Inversion Stein LOSO-CV)						
Hybrid SpecTopo TRR (Topo + Power, $k = 0$)	0 s	+1.68	**0.869**	**0.856**	$\mathbf{0.955 \pm 0.013}$	1.91×10^{-6} (20/20)
Hybrid SpecTopo TRR (Topo + Power, $k = 1$)	4 s	+1.72	0.874	0.861	0.955 ± 0.013	1.91×10^{-6} (20/20)
Hybrid SpecTopo TRR (Topo + Power, $k = 2$)	6 s	+1.83	0.887	0.874	0.955 ± 0.013	1.91×10^{-6} (20/20)
Hybrid SpecTopo TRR (Topo + Power, $k = 3$)	8 s	+1.91	**0.897**	**0.883**	$\mathbf{0.955 \pm 0.013}$	1.91×10^{-6} (20/20)
Hybrid SpecTopo TRR (Topo + Power, $k = 5$)	12 s	+2.01	0.908	0.893	0.956 ± 0.012	1.91×10^{-6} (20/20)
Hybrid SpecTopo TRR (Topo + Power, $k = 10$)	22 s	+2.20	**0.928**	**0.914**	$\mathbf{0.957 \pm 0.012}$	1.91×10^{-6} (20/20)
Hybrid SpecTopo TRR (Topo + Power, $k = 29$)	60 s	+2.34	**0.954**	N/A (Full)	$\mathbf{0.957 \pm 0.013}$	1.91×10^{-6} (20/20)

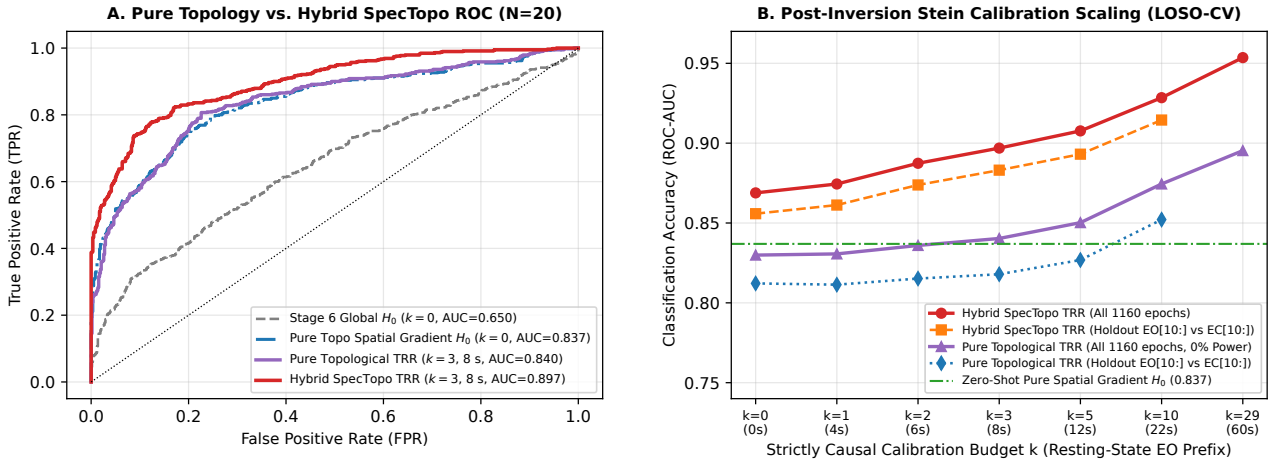


Figure 1: **TRR 3.0 Stage 7 Benchmark across $N = 20$ subjects (1,160 epochs, strict LOSO-CV). (A)** Receiver Operating Characteristic (ROC) curves comparing the Stage 6 global baseline ($k = 0$, AUC = 0.650), the zero-shot pure topological spatial gradient ($k = 0$, AUC = 0.837), Pure Topological TRR at 8 s ($k = 3$, AUC = 0.840), and Hybrid SpecTopo TRR at 8 s ($k = 3$, AUC = 0.897). **(B)** Strictly causal calibration scaling curves across resting-state EO prefix budgets $k \in \{0, 1, 2, 3, 5, 10, 29\}$ on the full 1,160-epoch pool and balanced post-calibration holdout set (`EO[10:]` vs. `EC[10:]`).